Yarn Dyeing Quality Control

Specification v2.0 — confirmed characteristic set
Ready for master data build

Plant	2002 · Dyeing
Characteristics	37 across three stage gates
Supersedes	v1.0 draft of 21 Aug 2026
Basis of change	QC Manager pruning of the v1 proposal
Backend	Standard SAP QM, inspection types 01 / 03 / 04
Next step	Five open decisions, then QS21 entry
Date	21 August 2026

> **This is the build specification.** The characteristic set is now yours, not my proposal. Sections 3–5 are written to be entered into `QS21` directly. Section 2 holds five decisions that must be settled first, and section 7 records what the pruning gave up, so the trade-offs stay visible rather than forgotten.

1. What changed from v1

Stage	v1 proposed	v2 confirmed	Net change
Stage 1 — Raw material	19	**9**	Ten cut. Three added that were not in v1: **Lot no.**, **Denier/Filaments**, **Grade**
Stage 2 — After dyeing	23	**15**	Twelve named, of which three split into six (see 2.1). Light fastness, metamerism, pH, tenacity retention cut
Stage 3 — After winding	18	**13**	Splice and Classimat tests cut. Three added: **Final Denier**, **Final Elongation**, **Oil %**
Total		**37**	Down from 60. A 38% reduction in master data and a materially lighter daily load

Three of your changes improve on my proposal

Denier and elongation measured at both ends. You kept elongation at stage 1 and added Final Denier and Final Elongation at stage 3. That is a better design than mine. It gives you a matched pair across the whole process, so the delta — not just the absolute value — becomes the signal. If elongation drops between greige and cone, the loss happened in your house and the data says where.

Oil % at stage 3. I had residual spin finish at stage 2, which is a scouring-efficiency check. Oil % after winding is a different and more commercially relevant measurement: it is the **coning oil** applied at the winder, and it governs how the yarn runs on the customer's knitting machine. Too little and it burns at the needle; too much and it stains. This belongs at stage 3 and I should have had it there.

Sublimation fastness retained. The one fastness test that is specific to disperse dyes on polyester, and the one most often forgotten. Keeping it while cutting hot pressing and water is the right order of priority.

2. Five decisions before QS21 entry

Q1. Should the composite fastness tests be split?
You named three: washing, rubbing, perspiration. Each is **rated on two separate scales with different acceptance limits**, and SAP QM records one result per characteristic. Rubbing dry passes at grade 4; rubbing wet passes at grade 3 — a single characteristic cannot hold both.

My recommendation: split into six. Washing → colour change + staining. Rubbing → dry + wet. Perspiration → acid + alkaline. Tables in section 4 assume this. Say the word and I will collapse them to three, but then only one limit per test can be enforced automatically.

Q2. What is "Grade"?
Supplier-declared yarn grade taken from the delivery document (AA / A / B), or a grade your own QC assigns after inspection?

- **If supplier-declared** — it is context, not a test result. It belongs on the app header, read from the batch, and needs no MIC.
- **If QC-assigned** — it is a genuine qualitative characteristic with a code list, and I need the grade definitions to build the selected set.

Section 3 currently assumes **QC-assigned**, which is the more useful of the two.

Q3. Lot no. and Denier/Filaments — recorded, or verified?
These are attributes that already exist in your system rather than things a test produces. My proposal:

- **Lot no.** → the supplier lot goes in `QALS-LICHA` (vendor batch), a standard field SAP already provides. Shown and editable on the app header. Not an MIC.
- **Denier / Filaments** → displayed on the header from `ZZMARA.ZZDENIR` and `ZZFILAM`, **and** denier separately *verified* by measurement as MIC `S1_DEN` against ISO 2060. Filament count is verified visually only where there is doubt.

This gives you both the record and the check. Confirm or correct.

Q4. Free formaldehyde and banned azo amines — in-house or external?
Both are normally sent to an accredited laboratory, and azo amine testing to EN ISO 14362-1 effectively always is. If external, the app captures the certificate result and reference number rather than prompting for a live measurement, and the tier changes from per-batch to per-certification-cycle. This changes the screen design, so it needs deciding now.

Q5. What frequency for stage 1?
Stage 1 is per **supplier delivery lot**, not per dye batch — the same greige lot feeds many batches. Roughly how many greige deliveries arrive per day or per week? The number drives the daily volume in section 6 and the sampling plan in section 3.4.

3. Stage 1 — Raw material, before dyeing

Per supplier delivery lot. Inspection type **01**, goods receipt against purchase order. Anchor: `ZPP_BATCHN.GREY_CODE` + vendor batch.

3.1 Header context — displayed, not tested

Field	Source	Behaviour in the app
Lot no.	`QALS-LICHA` (vendor batch)	Scanned or typed at lot start; carried on every result
Denier / Filaments	`ZZMARA.ZZDENIR` / `ZZFILAM`	Read-only on header; denier also verified as `S1_DEN` below
Material / SAP batch	`ZPP_BATCHN.GREY_CODE`	Read-only; the reference for `S1_IDT`

3.2 Characteristics for QS21

MIC	Characteristic	Standard	Type	Unit	Proposed limit	Tier	Method note
S1_DEN	Linear density (denier verification)	ISO 2060 ASTM D1907	Quant	dtex	nominal ±2.0%	1	Skein, 100 m wraps, conditioned 20°C / 65% RH
S1_ELG	Elongation at break	ISO 2062 ASTM D2256	Quant	%	18–28	1	CRE, 500 mm gauge, 20 breaks, report mean
S1_BWS	**Boiling water shrinkage**	ASTM D2259	Quant	%	5.0–9.0 **range ≤ 1.5**	1	100°C / 30 min. Record on 5 packages — **the spread matters more than the mean**
S1_BFL	Broken filaments	Visual under tension	Quant	count/kg	≤ 2	1	10-minute run over a black board at constant tension
S1_PKD	Package density	Mass / volume	Quant	g/cm³	0.36–0.42	1	**Governs liquor flow.** Hard packages dye light in the core
S1_PKW	Package weight	—	Quant	kg	nominal ±3%	1	Sets the dyebath liquor ratio
S1_GRD	Grade	Internal — *see Q2*	Qual	code	AA or A	1	Code list required before this can be built
S1_IDT	Lot identity vs SAP batch	—	Qual	P/F	Pass	1	Supplier tag vs `GREY_CODE`. **Single-lot loading rule — see 3.3**

3.3 The single-lot rule

Loading one dye vessel from two supplier lots is among the most reliable ways to produce barré in polyester, because dye affinity differs between lots in ways no dyehouse process control can correct. Your *Merge Details* app suggests merging is a recognised operation here.

Recommendation: make `S1_IDT` fail automatically when a batch draws on more than one vendor batch, and require a supervisor override with a recorded reason. This is one rule, cheap to implement, and it protects the largest single loss mode at stage 2.

3.4 Sampling

Characteristic	Sample	Positioning
`S1_DEN`, `S1_ELG`	20 breaks across 5 packages	Spread across the delivery, not from one carton
`S1_BWS`	5 packages	Different cartons — the point is to expose lot spread
`S1_PKD`, `S1_PKW`	ISO 2859-1 GIL II	See 6.2 for the sample-size table
`S1_BFL`, `S1_GRD`, `S1_IDT`	ISO 2859-1 GIL II, AQL 2.5 / 1.0	Result recorded as **number of defectives found**, one entry per lot

4. Stage 2 — After dyeing

Per dye batch, after dyeing and drying, before release to winding. Inspection type **03**, in-process. Anchor: `ZPP_BATCHN.DYE_CODE` + `AUFNR`.

4.1 Shade and appearance — every batch

MIC	Characteristic	Standard	Type	Unit	Proposed limit	Tier	Method note
`S2_DE`	**Shade vs approved standard**	ISO 105-J03 AATCC EP6	Quant	ΔE CMC(2:1)	≲ 1.0	1	Spectrophotometer D65/10°, specular included, UV calibrated. Wind a card to constant thickness
`S2_INOUT`	**Inside–outside package variation**	ISO 105-A02	Qual	grade	≳ 4-5	1	Innermost vs outermost 50 g of the same cheese. **The failure mode package dyeing owns**
`S2_P2P`	Package-to-package within batch	ISO 105-A02	Qual	grade	≳ 4	1	Top, middle, bottom of the carrier — exposes vertical flow gradient
`S2_B2B`	Batch-to-batch vs master	ISO 105-A02	Qual	grade	≳ 4	1	Against the retained master for that shade
`S2_LEVEL`	Levelness / barré	ISO 105-A02	Qual	grade	≳ 4	1	Knitted sock, D65 light cabinet
`S2_STAIN`	Stains, tip-dyeing, rust	Visual	Qual	P/F	Pass	1	Defect code required on fail. Tip-dyeing points at carrier or flow problems

4.2 Colour fastness — per recipe qualification (assumes Q1 answered "split")

MIC	Characteristic	Standard	Type	Unit	Proposed limit	Tier	Method note
`S2_WFC`	Washing — colour change	ISO 105-C06 A1S	Qual	grade	≳ 4	2	Grey scale ISO 105-A02
`S2_WFS`	Washing — staining	ISO 105-C06 A1S	Qual	grade	≳ 3-4	2	Multifibre adjacent, grey scale ISO 105-A03
`S2_RUBD`	Rubbing — dry	ISO 105-X12	Qual	grade	≳ 4	2	Crockmeter
`S2_RUBW`	Rubbing — wet	ISO 105-X12	Qual	grade	≳ 3	2	**Most frequent fastness failure in deep shades**
`S2_PSPA`	Perspiration — acid	ISO 105-E04	Qual	grade	≳ 4	2	Perspirometer
`S2_PSPB`	Perspiration — alkaline	ISO 105-E04	Qual	grade	≳ 4	2	Perspirometer
`S2_SUBL`	**Sublimation / dry heat**	ISO 105-P01	Qual	grade	≳ 4	2	180°C / 30 s. Specific to disperse dyes — the one not to skip

4.3 Chemical compliance (tier depends on Q4)

MIC	Characteristic	Standard	Type	Unit	Proposed limit	Tier	Method note
`S2_HCHO`	Free formaldehyde	ISO 14184-1	Quant	ppm	≲ 75	3	OEKO-TEX Class II. Certificate reference captured if external
`S2_AZO`	Banned azo amines	EN ISO 14362-1	Quant	ppm	< 20	3	REACH Annex XVII. Almost always an external accredited lab

> **Fastness is a recipe property, not a batch property.** Two batches of the same recipe, run correctly, will not differ in fastness. Qualifying each recipe once — and re-qualifying on any change of recipe, dye lot or dye supplier — is scientifically sound and holds up in a buyer audit, provided the link from batch to qualified recipe is recorded. Guaranteeing that link is the app's job, and it is why Q5 in v1 about recipe identification still matters.

5. Stage 3 — After winding

Per wound lot, before packing. Inspection type **04**, GR from production. Anchor: `ZPP_PACK` / handling unit.

5.1 Package geometry and build

MIC	Characteristic	Standard	Type	Unit	Proposed limit	Tier	Method note
`S3_PKW`	Package weight	—	Quant	kg	nominal ±1.5%	1	Net. Commercial exposure — short weight is a claim
`S3_PKL`	Package length	ISO 2060	Quant	m	nominal ±2%	2	From the winder metering counter, verified periodically
`S3_PKD`	Package density	Mass / volume	Quant	g/cm³	0.42–0.50	1	Soft collapses in transit, hard will not unwind
`S3_TAPER`	Cone taper and traverse	—	Qual	P/F	Pass	1	Gauge against the agreed cone format
`S3_RIBB`	Ribboning / patterning	Visual	Qual	P/F	Pass	1	Causes sloughing at the customer's creel
`S3_NOSE`	Nose bulging	Visual	Qual	P/F	Pass	1	Also check soft and hard edges while handling

5.2 Yarn condition after winding

MIC	Characteristic	Standard	Type	Unit	Proposed limit	Tier	Method note
`S3_BFL`	Broken filaments	Visual under tension	Quant	count/kg	≲ 2	1	**Compare with `S1_BFL` — the delta is what your process added**
`S3_TENS`	Winding tension	Tensiometer	Quant	cN	target ±10% CV ≲ 6%	2	Per winding position. Doubles as a machine-maintenance signal
`S3_DEN`	**Final denier**	ISO 2060	Quant	dtex	nominal ±2.0%	2	Paired with `S1_DEN`
`S3_ELG`	**Final elongation**	ISO 2062	Quant	%	18–28 **loss vs S1 ≲ 8%**	2	Paired with `S1_ELG`. The loss figure detects thermal or mechanical damage

5.3 Final release

MIC	Characteristic	Standard	Type	Unit	Proposed limit	Tier	Method note
`S3_CONT`	Contamination / stain	Visual	Qual	P/F	Pass	1	Oil from the winder, rust, foreign fibre. Defect code on fail
`S3_MOIS`	Moisture	ISO 6741-1	Quant	%	≲ 0.5	1	Oven dry 105°C to constant mass, or calibrated moisture balance
`S3_OIL`	**Oil % (coning oil)**	Solvent extraction	Quant	%	1.0–2.0 *confirm target*	1	**Applied at the winder, not residual spin finish.** Governs running on the customer's knitting machine

> `S3_OIL` **needs your target, not mine.** Coning oil pick-up is set by what the yarn is for — knitting yarn typically carries more than weaving yarn, and different customers specify differently. The 1.0–2.0% above is a placeholder. Please replace it with your own applicator setting and tolerance before this goes into master data.

6. Volume, and what it means for the apps

6.1 Recalculated daily load

Stage	Frequency	Tier 1 chars	Events / day	Result entries / day
Stage 1	Per delivery lot	8	*pending Q5* — assume 6	~48
Stage 2 tier 1	Every batch	6	75	**450**
Stage 2 tier 2	Per recipe	7	~10 new recipes	~70
Stage 3 tier 1	Every wound lot	9	75	**675**
Stage 3 tier 2	Periodic	4	~15	~60
Total				**~1,300 result entries per day**

> **That is about two hours of data entry across the whole day**, at four to five seconds per result, split between the laboratory and the winding floor. It is higher than the 625 in v1 because stage 3 turned out to be nine tier-1 characteristics rather than my assumed lighter set. Two hours spread across two locations and a full shift is workable, but it is the number the app design has to earn — and it is why section 6.2 matters more than it looks.

6.2 One design rule that makes this possible

> **Attribute characteristics are recorded once per lot, not once per package.**
>
> Under ISO 2859-1 you inspect a sample of *n* packages and record **the number of defectives found**. So `S3_RIBB` on a 32-package sample is **one** result — "2 defectives" — not thirty-two pass/fail entries.
>
> Get this wrong and stage 3 alone becomes 2,400 entries a day and the system is abandoned within a month. The apps must present a single counter per attribute characteristic, pre-loaded with the sample size derived from the cheese or cone count. This is the most important single decision in the build.

6.3 Sampling table — ISO 2859-1, GIL II, normal severity

Lot size (packages)	Code letter	Sample size	Ac / Re at AQL 2.5	Ac / Re at AQL 1.0
26 – 50	D	8	0 / 1	0 / 1
51 – 90	E	13	1 / 2	0 / 1
91 – 150	F	20	1 / 2	1 / 2
151 – 280	G	32	2 / 3	1 / 2
281 – 500	H	50	3 / 4	1 / 2

AQL 1.0 for critical (identity, chemical compliance) · AQL 2.5 for major (shade, package integrity, broken filaments) · AQL 4.0 for cosmetic. Lot size comes from `ZPP_BATCHN.CHEESES` or the packing record, so the app derives the sample size itself.

7. Risk register — what the pruning gave up

Recorded so the trade-offs stay visible. None of these is an argument to reverse your decision; they are the things to watch, and the first places to look when something goes wrong.

Cut from v1	What it protected against	Risk	Residual mitigation in v2
Dye uniformity tube test (stage 1)	Package-to-package dye affinity variation — the root cause of barré that no dyehouse control can correct	**High**	**Partial, and better than nothing.** `S1_BWS` is retained, and BWS spread across packages is a recognised proxy for dye affinity spread. I have therefore added a **range ≲ 1.5%** limit alongside the absolute range. Watch this figure — if barré rises at stage 2, the tube test is the first thing to reinstate
Tenacity (stages 1 and 2)	Yarn strength and dyebath damage	Medium	`S3_ELG` with the loss-vs-`S1_ELG` limit catches gross damage. Elongation loss and tenacity loss usually move together
Light fastness ISO 105-B02	Fading in service — a common cause of customer claims on furnishing and automotive yarn	Medium	None. If any customer specifies light fastness, it must be reinstated for those recipes and outsourced
Splice strength and count	Yarn breaks at the customer's machine	Medium	None. `S3_TENS` gives an indirect signal only. Worth revisiting if running complaints appear
pH ISO 3071 (stage 2)	Incomplete wash-off; skin-contact compliance	Low–Med	None. Cheap and fast — the easiest of these to add back if OEKO-TEX certification is pursued
Package hardness (stage 1)	Core-to-surface dyeing variation	Low	`S1_PKD` density is retained and measures much the same property
Label / barcode verification	Wrong yarn shipped under the right label	Low	None in QM. Likely already covered by handling-unit scanning in the packing process

8. Master data build list

Object	Count	Transaction	Content
Master inspection characteristics	**37**	`QS21`	8 stage 1, 15 stage 2, 13 stage 3, plus `S1_GRD` pending Q2
— quantitative with limits	19		Limits from sections 3–5
— qualitative grey scale	7		Nine-code selected set
— qualitative pass/fail	10		With defect catalogs
— qualitative coded	1		`S1_GRD` — blocked on Q2
Code groups / catalogs	5	`QS41`	Grey scale, pass/fail, grade, plus defect catalogs for stages 2 and 3
Selected sets	5	`QS51`	With per-characteristic valuation thresholds
Inspection plans	3	`QP01`	One per stage, per material group
Sampling procedures	3	`QDV1`	ISO 2859-1 attribute, fixed-sample variable, 100% inspection
Dynamic modification rules	1	`QDR1`	Implements tier-2 skipping once a recipe is qualified
Usage decision selected set	1	`QS51`	Catalog 3 — A, A1, RD, ST, DG, R

> **Down from ~58 MICs to 37.** Roughly two days of entry rather than three, but the saving that matters is in ongoing operation, not in setup. Every characteristic you keep is a characteristic somebody stands at a bench and measures, 75 times a day, for years.

9. Next steps

1. **Answer Q1–Q5** in section 2. Q1 (fastness splitting) and Q2 (grade definition) block master data entry; the others shape the app.
2. **Confirm the `S3_OIL` target** from your winder applicator setting.
3. **Review the limits.** Every figure in sections 3–5 is an industry-typical proposal. Check each against your own process capability — a limit the plant cannot meet teaches operators to ignore the system, which is worse than having no limit at all.
4. **Master data build** once 1–3 are settled. QM consultant, roughly two days plus the review sessions.

5. **Build Post-Dyeing QC** end to end as the pattern app — 15 characteristics, the fullest of the three, and the one that exercises grey scale, pass/fail, quantitative and external-certificate capture together.
6. **Pilot two weeks** on one dyeing line before replicating to stages 1 and 3.

Specification v2.0. Supersedes v1.0 of 21 August 2026. The characteristic set is as confirmed by the QC Manager; standards, methods, tiers and limits are proposed against those characteristics and remain subject to review under step 3 above.